

Endogenous Grid Method (EGM)

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Recap: Households' Dynamic Optimization Problem

- ▶ The representative household chooses $\{c_t, k_{t+1}\}$ to maximize lifetime utility:

$$\max_{\{c_t, k_{t+1}\}_{t \geq 0}} \sum_{t=0}^{\infty} \beta^t u(c_t) \quad \text{s.t.} \quad c_t + k_{t+1} = R_t k_t + w_t, \quad k_{t+1} \geq b.$$

where $R_t = 1 + r_t$ and $P(w_{t+1}|w_t)$ is taken as given to agent

- ▶ State k_t, w_t determine the income
- ▶ Controls c_t, k_{t+1} determine the expenditure. We pick $c_t = \pi(k_t, w_t)$ and then $k_{t+1} = R_t k_t + w_t - \pi(k_t, w_t)$
- ▶ Recursively (drop index t), we derived the Euler equation in the case of not binding borrowing constraint:

$$u_c(c) = \beta R \mathbb{E}[u_c(c')] = \beta R \int_{w'} u_c(c') P(dw' | w) \quad \text{if } k' \geq b$$

Recap: Dynamic Optimization as a KKT Problem

- ▶ $k' = Rk + w - \pi(k, w) \geq b$ can be rewritten as $\pi(k, w) \leq Rk + w - b \equiv \bar{\pi}(k, w)$.
- ▶ Replacing c with $\pi(k, w)$ and inverting u_c gives

$$\begin{aligned} \pi(k, w) = \min \left\{ u_c^{-1} \left[\beta R \int_{w'} u_c(\pi(k', w')) P(dw' | w) \right], \bar{\pi}(k, w) \right\} \\ \text{s.t. } k' = Rk + w - \pi(k, w). \end{aligned}$$

The policy π^* that satisfies the above is the solution.

- ▶ Time Iteration (TI): define the Coleman operator H so that, for a policy $\pi^{(i)}$,

$$\begin{aligned} \pi^{(i+1)}(k, w) = H(\pi^{(i)})(k, w) = \min \left\{ u_c^{-1} \left[\beta R \int_{w'} u_c(\pi^{(i)}(k', w')) P(dw' | w) \right], \bar{\pi}(k, w) \right\} \\ \text{s.t. } k' = Rk + w - \pi^{(i+1)}(k, w). \end{aligned}$$

The fixed point $\pi^* = H(\pi^*)$ is the desired policy.

- ▶ For a particular (k, w) , $\pi^{(i+1)}(k, w)$ is solved implicitly because $k' = Rk + w - \pi^{(i+1)}(k, w)$ feeds back into the RHS of the Euler equation.

Time Iteration (TI): Implementation

- ▶ Represent the policy on a tensor grid: $\bar{\mathbf{k}} = \{\bar{k}_1, \dots, \bar{k}_M\}$ and $\bar{\mathbf{w}} = \{\bar{w}_1, \dots, \bar{w}_N\}$. Store the policy function $\pi^{(i)}$ in each iteration as a consumption array $\mathbf{c}^{(i)} \in \mathbb{R}^{M \times N}$ with $\pi^{(i)}(\bar{k}_m, \bar{w}_n) := c_{m,n}^{(i)}$.

- ▶ The saving array $\mathbf{k}'^{(i)}$ can be obtained by $k'_{m,n} = R \bar{k}_m + \bar{w}_n - c_{m,n}^{(i)}$
- ▶ View the policy as parameterized by the grids and coefficients:

$$\pi^{(i)}(k, w) \equiv \pi^{(i)}(k, w; \bar{\mathbf{k}}, \bar{\mathbf{w}}, \mathbf{c}^{(i)}).$$

- ▶ For $\bar{k}_m \leq k \leq \bar{k}_{m+1}$ at fixed $\bar{w}_n$, use linear interpolation

$$\lambda(k) := \frac{k - \bar{k}_m}{\bar{k}_{m+1} - \bar{k}_m} \in [0, 1], \quad \pi^{(i)}(k, \bar{w}_n) = (1 - \lambda(k)) c_{m,n}^{(i)} + \lambda(k) c_{m+1,n}^{(i)}.$$

This extends to bilinear interpolation over (k, w) (e.g., midpoints in w) and to higher-order schemes for smoothness.

Time Iteration (TI): Implementation

- ▶ The Coleman operator $H : \mathbb{R}^{M \times N} \rightarrow \mathbb{R}^{M \times N}$ maps $\mathbf{c}^{(i)}$ to $\mathbf{c}^{(i+1)} = H(\mathbf{c}^{(i)})$ by solving a nonlinear $M \times N$ -dimensional system of implicit scalar equations (one per $(\bar{k}_m, \bar{w}_n)$):

$$c_{m,n}^{(i+1)} = \min \left\{ u_c^{-1} \left(\beta R \sum_{n'=1}^N P_{nn'} u_c \left(\pi^{(i)}(k'_{m,n}{}^{(i+1)}, \bar{w}_{n'}; \bar{\mathbf{k}}, \bar{\mathbf{w}}, \mathbf{c}^{(i)}) \right) \right), \bar{\pi}_{m,n} \right\},$$
$$k'_{m,n}{}^{(i+1)} := R \bar{k}_m + \bar{w}_n - c_{m,n}^{(i+1)},$$
$$\bar{\pi}_{m,n} := R \bar{k}_m + \bar{w}_n - b.$$

- ▶ Here w' lives on the same grid $\bar{\mathbf{w}}$; the sum uses the next-period index n' with transition weights $P_{nn'}$ (from $\bar{w}_n$ to $\bar{w}_{n'}$).
- ▶ This is a nonlinear implicit equation for $c_{m,n}^{(i+1)}$: the unknown appears on both sides; the update is also piecewise due to the min with $\bar{\pi}_{m,n}$.
- ▶ $\pi^{(i)}(k'_{m,n}{}^{(i+1)}, \bar{w}_{n'})$ is evaluated by interpolating over the k -grid $\bar{\mathbf{k}}$ at each fixed $\bar{w}_{n'}$.

Endogenous Grid Method (EGM)

- ▶ Keep the grids $\bar{\mathbf{k}} = \{\bar{k}_1, \dots, \bar{k}_M\}$ and $\bar{\mathbf{w}} = \{\bar{w}_1, \dots, \bar{w}_N\}$ and the parameterization $\pi^{(i)}(k, w; \bar{\mathbf{k}}, \bar{\mathbf{w}}, \mathbf{c}^{(i)})$, where $\bar{k}_1 = b$
- ▶ Instead of treating $(\bar{\mathbf{k}}, \bar{\mathbf{w}})$ as today's states and inferring k' forward, treat them as tomorrow's states (so $k'_{m'} = \bar{k}_{m'}$) and infer current k backward via the Euler equation:

$$\tilde{c}_{m',n}^{(i+1)} = u_c^{-1} \left(\beta R \sum_{n'=1}^N P_{nn'} u_c \left(\pi^{(i)}(\bar{k}_{m'}, \bar{w}_{n'}; \bar{\mathbf{k}}, \bar{\mathbf{w}}, \mathbf{c}^{(i)}) \right) \right).$$

The right-hand side is explicit (no root finding) because all inputs are known.

- ▶ We do not apply the borrowing cap here since $\bar{\pi}(k, w) = Rk + w - b$ depends on the current k , which is not yet known; this will be enforced after we back out $\tilde{k}$ and interpolate onto the current- k grid.
- ▶ Here $\tilde{c}_{m',n}^{(i+1)}$ is the unconstrained consumption at off-grid point $\tilde{k}_{m',n}$; it is not yet the desired on-grid value $c_{m,n}^{(i+1)}$.
- ▶ Recover the associated current state from the budget constraint:

$$\bar{k}_{m'} = R \tilde{k}_{m',n}^{(i+1)} + \bar{w}_n - \tilde{c}_{m',n}^{(i+1)} \implies \tilde{k}_{m',n}^{(i+1)} = \frac{\bar{k}_{m'} - \bar{w}_n + \tilde{c}_{m',n}^{(i+1)}}{R}.$$

EGM: Interpolation

- ▶ Recall that the target parameterization is $\pi^{(i+1)}(k, w; \bar{\mathbf{k}}, \bar{\mathbf{w}}, \mathbf{c}^{(i+1)})$, while the current (off-grid) parameterization is $\pi^{(i+1)}(k, w; \tilde{\mathbf{k}}^{(i+1)}, \tilde{\mathbf{w}}, \tilde{\mathbf{c}}^{(i+1)})$.
- ▶ We apply the interpolation scheme described earlier: for each grid point $\bar{w}_n$ (outer loop) and each grid point $\bar{k}_m$ (inner loop):
 - ▶ find adjacent points in $\tilde{\mathbf{k}}^{(i+1)}$ such that $\tilde{k}_{m',n}^{(i+1)} \leq \bar{k}_m \leq \tilde{k}_{m'+1,n}^{(i+1)}$, then compute the weight

$$\lambda(\bar{k}_m) := \frac{\bar{k}_m - \tilde{k}_{m',n}^{(i+1)}}{\tilde{k}_{m'+1,n}^{(i+1)} - \tilde{k}_{m',n}^{(i+1)}} \in [0, 1]$$

and therefore

$$\pi^{(i+1)}(\bar{k}_m, \bar{w}_n) := c_{m,n}^{(i+1)} = (1 - \lambda(\bar{k}_m)) \tilde{c}_{m',n}^{(i+1)} + \lambda(\bar{k}_m) \tilde{c}_{m'+1,n}^{(i+1)}.$$

- ▶ if $\bar{k}_m < \tilde{k}_{1,n}^{(i+1)}$: $\pi^{(i+1)}(\bar{k}_m, \bar{w}_n) = \bar{\pi}_{m,n}$ where $\bar{\pi}_{m,n} := R \bar{k}_m + \bar{w}_n - b$
- ▶ if $\bar{k}_m > \tilde{k}_{M',n}^{(i+1)}$: use monotone linear extrapolation from $(\tilde{k}_{M'-1,n}^{(i+1)}, \tilde{c}_{M'-1,n}^{(i+1)})$ and $(\tilde{k}_{M',n}^{(i+1)}, \tilde{c}_{M',n}^{(i+1)})$, or clamp: set $\pi^{(i+1)}(\bar{k}_m, \bar{w}_n) = \tilde{c}_{M',n}^{(i+1)}$

Time Iteration (TI): Pseudocode (1/2)

1. Prerequisites:

- ▶ Utility $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$, marginal utility $u_c(c) = c^{-\sigma}$.
- ▶ Parameters β, R, σ, b ; transition matrix $P = (P_{nn'})$.
- ▶ Grids: $\bar{\mathbf{k}} = \{\bar{k}_1, \dots, \bar{k}_M\}$ with $\bar{k}_1 = b$, and $\bar{\mathbf{w}} = \{\bar{w}_1, \dots, \bar{w}_N\}$.
- ▶ Solver hyperparameters: MaxIter, tolerance ε .

2. Initialize:

- ▶ Borrowing cap (elementwise) $\bar{\pi}_{m,n} := R\bar{k}_m + \bar{w}_n - b$.
- ▶ Set $c_{m,n}^{(0)} = \bar{\pi}_{m,n}$ for all (m,n) . This defines $\pi^{(0)}(k, w; \bar{\mathbf{k}}, \bar{\mathbf{w}}, \mathbf{c}^{(0)})$.

3. Iterate for $i = 0, 1, \dots, \text{MaxIter} - 1$:

- ▶ For each node (m, n) , define the residual and an Nonlinear Complementarity Problem (NCP) reformulation:

$$G_{m,n}(c) := u_c(c) - \beta R \sum_{n'=1}^N P_{nn'} u_c\left(\pi^{(i)}(R\bar{k}_m + \bar{w}_n - c, \bar{w}_{n'}; \bar{\mathbf{k}}, \bar{\mathbf{w}}, \mathbf{c}^{(i)})\right),$$

$$\Psi_{m,n}(c) := \sqrt{G_{m,n}(c)^2 + (\bar{\pi}_{m,n} - c)^2} - G_{m,n}(c) - (\bar{\pi}_{m,n} - c).$$

Time Iteration (TI): Pseudocode (2/2)

3. Iterate (continued):

- ▶ Solve for $c_{m,n}^{(i+1)} \in [0, \bar{\pi}_{m,n}]$ using either:
 - ▶ Bracketing (bisection/Brent) on $G_{m,n}(c) = 0$ with clipping at $\bar{\pi}_{m,n}$, or
 - ▶ Semismooth Newton on the Fischer–Burmeister equation $\Psi_{m,n}(c) = 0$.
- ▶ Set $k'_{m,n} = R \bar{k}_m + \bar{w}_n - c_{m,n}^{(i+1)}$.

4. Convergence:

- ▶ If $\max_{m,n} |c_{m,n}^{(i+1)} - c_{m,n}^{(i)}| < \varepsilon$, stop.
- ▶ Otherwise, form $\pi^{(i+1)}(k, w; \bar{\mathbf{k}}, \bar{\mathbf{w}}, \mathbf{c}^{(i+1)})$ and continue.

Euler Residual Function

- Define the node-wise Euler residual (with $k' = R\bar{k}_m + \bar{w}_n - c$):

$$G_{m,n}(c) := u_c(c) - \beta R \sum_{n'=1}^N P_{nn'} u_c\left(\pi^{(i)}(k', \bar{w}_{n'}; \bar{\mathbf{k}}, \bar{\mathbf{w}}, \mathbf{c}^{(i)})\right).$$

- Monotonicity (unique root): since $u_{cc} < 0$ and $k' = R\bar{k}_m + \bar{w}_n - c$,

$$\frac{dG_{m,n}}{dc} = u_{cc}(c) + \beta R \sum_{n'} P_{nn'} u_{cc}\left(\pi^{(i)}(k', \bar{w}_{n'})\right) \frac{\partial \pi^{(i)}(k', \bar{w}_{n'})}{\partial k'} < 0,$$

where $\partial \pi^{(i)} / \partial k' \geq 0$ by monotonicity in resources, and $u_{cc} < 0$. Hence $G_{m,n}$ is strictly decreasing $\Rightarrow$ at most one root.

- Let c^* be the unconstrained root, $G_{m,n}(c^*) = 0$. The feasible set is $[0, \bar{\pi}_{m,n}]$ with $\bar{\pi}_{m,n} := R\bar{k}_m + \bar{w}_n - b$.
 - If $c^* \leq \bar{\pi}_{m,n}$: interior solution $c_{m,n}^{(i+1)} = c^*$.
 - If $c^* > \bar{\pi}_{m,n}$: boundary solution $c_{m,n}^{(i+1)} = \bar{\pi}_{m,n}$.

Complementarity (NCP) formulations

- ▶ KKT at node (m, n) (an NCP in c):

$$G_{m,n}(c_{m,n}^{(i+1)}) \geq 0 \perp \bar{\pi}_{m,n} - c_{m,n}^{(i+1)} \geq 0.$$

- ▶ Use the min operator to unify the interior and binding cases (single-equation NCP):

$$\min\{G_{m,n}(c), \bar{\pi}_{m,n} - c\} = 0.$$

- ▶ Fischer–Burmeister (FB) reformulation (single equation, differentiable almost everywhere):

$$\Psi_{m,n}(c) := \sqrt{G_{m,n}(c)^2 + (\bar{\pi}_{m,n} - c)^2} - G_{m,n}(c) - (\bar{\pi}_{m,n} - c) = 0.$$

- ▶ Practical note: the FB map works well with autodiff when we later introduce machine-learning-based solvers.

Endogenous Grid Method (EGM): Pseudocode (1/2)

1. Prerequisites:

- ▶ Utility $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$, marginal utility $u_c(c) = c^{-\sigma}$.
- ▶ Parameters β, R, σ, b ; transition matrix $P = (P_{nn'})$.
- ▶ Grids: $\bar{\mathbf{k}} = \{\bar{k}_1, \dots, \bar{k}_M\}$ with $\bar{k}_1 = b$, and $\bar{\mathbf{w}} = \{\bar{w}_1, \dots, \bar{w}_N\}$.
- ▶ Solver hyperparameters: MaxIter, tolerance ε .

2. Initialize:

- ▶ Borrowing cap (elementwise) $\bar{\pi}_{m,n} := R \bar{k}_m + \bar{w}_n - b$.
- ▶ Set $c_{m,n}^{(0)} = \bar{\pi}_{m,n}$ for all (m,n) . This defines $\pi^{(0)}(k, w; \bar{\mathbf{k}}, \bar{\mathbf{w}}, \mathbf{c}^{(0)})$.

3. Iterate for $i = 0, 1, \dots, \text{MaxIter} - 1$:

- ▶ Backward consumption for $\tilde{\mathbf{c}}^{(i+1)}$. For each (m', n) :

$$\tilde{c}_{m',n}^{(i+1)} = u_c^{-1} \left(\beta R \sum_{n'=1}^N P_{nn'} u_c \left(\pi^{(i)}(\bar{k}_{m'}, \bar{w}_{n'}; \bar{\mathbf{k}}, \bar{\mathbf{w}}, \mathbf{c}^{(i)}) \right) \right).$$

- ▶ Implied current state $\tilde{\mathbf{k}}$. For each (m', n) : $\tilde{k}_{m',n}^{(i+1)} = \frac{\bar{k}_{m'} - \bar{w}_n + \tilde{c}_{m',n}^{(i+1)}}{R}$.

Endogenous Grid Method (EGM): Pseudocode (2/2)

3. Iterate (continued):

► Interpolation. For each (m, n) :

- If $\bar{k}_m < \tilde{k}_{1,n}^{(i+1)}$: set $c_{m,n}^{(i+1)} = \bar{\pi}_{m,n}$
- Else if $\bar{k}_m > \tilde{k}_{M,n}^{(i+1)}$: use monotone linear extrapolation from the last two pairs $(\tilde{k}_{M-1,n}^{(i+1)}, \tilde{c}_{M-1,n}^{(i+1)})$, $(\tilde{k}_{M,n}^{(i+1)}, \tilde{c}_{M,n}^{(i+1)})$, or clamp to $c_{m,n}^{(i+1)} = \tilde{c}_{M,n}^{(i+1)}$.
- Else (interior): find m' s.t. $\tilde{k}_{m',n}^{(i+1)} \leq \bar{k}_m \leq \tilde{k}_{m'+1,n}^{(i+1)}$, set

$$\lambda(\bar{k}_m) = \frac{\bar{k}_m - \tilde{k}_{m',n}^{(i+1)}}{\tilde{k}_{m'+1,n}^{(i+1)} - \tilde{k}_{m',n}^{(i+1)}}, \quad c_{m,n}^{(i+1)} = (1 - \lambda) \tilde{c}_{m',n}^{(i+1)} + \lambda \tilde{c}_{m'+1,n}^{(i+1)}.$$

- (Optional) Enforce cap: $c_{m,n}^{(i+1)} \leftarrow \min\{c_{m,n}^{(i+1)}, \bar{\pi}_{m,n}\}$.
- (Optional) Damping: $c^{(i+1)} \leftarrow (1 - \alpha)c^{(i)} + \alpha c^{(i+1)}$, with $\alpha \in (0, 1]$.
- set $k'_{m,n} = R\bar{k}_m + \bar{w}_n - c_{m,n}^{(i+1)}$

4. Convergence:

- If $\max_{m,n} |c_{m,n}^{(i+1)} - c_{m,n}^{(i)}| < \varepsilon$, stop.
- Otherwise, form $\pi^{(i+1)}(k, w; \bar{\mathbf{k}}, \bar{\mathbf{w}}, \mathbf{c}^{(i+1)})$ and continue.

Forward rollout (lagged-Coleman explicit update): Pseudocode (1/2)

1. Prerequisites:

- ▶ Utility $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$, marginal utility $u_c(c) = c^{-\sigma}$.
- ▶ Parameters β, R, σ, b ; transition matrix $P = (P_{nn'})$.
- ▶ Grids: $\bar{\mathbf{k}} = \{\bar{k}_1, \dots, \bar{k}_M\}$ with $\bar{k}_1 = b$, and $\bar{\mathbf{w}} = \{\bar{w}_1, \dots, \bar{w}_N\}$.
- ▶ Solver hyperparameters: MaxIter, tolerance ε .

2. Initialize:

- ▶ Borrowing cap (elementwise) $\bar{\pi}_{m,n} := R \bar{k}_m + \bar{w}_n - b$.
- ▶ Set $c_{m,n}^{(0)} = \bar{\pi}_{m,n}$ for all (m,n) . This defines $\pi^{(0)}(k, w; \bar{\mathbf{k}}, \bar{\mathbf{w}}, \mathbf{c}^{(0)})$.

Forward rollout (lagged-Coleman explicit update): Pseudocode (2/2)

3. Iterate for $i = 0, 1, \dots, \text{MaxIter} - 1$:

- ▶ For each node (m, n) , form the next state using the **old policy (explicit rollout)**:

$$k'_{m,n}{}^{(i)} := R \bar{k}_m + \bar{w}_n - c_{m,n}^{(i)}.$$

- ▶ Compute the unconstrained updated consumption via the Euler RHS under $\pi^{(i)}$:

$$\tilde{c}_{m,n}^{(i+1)} = u_c^{-1} \left(\beta R \sum_{n'=1}^N P_{nn'} u_c \left(\pi^{(i)}(k'_{m,n}, \bar{w}_{n'}; \bar{\mathbf{k}}, \bar{\mathbf{w}}, \mathbf{c}^{(i)}) \right) \right).$$

- ▶ Enforce feasibility on today's grid (borrowing cap):

$$c_{m,n}^{(i+1)} = \min \{ \tilde{c}_{m,n}^{(i+1)}, \bar{\pi}_{m,n} \}.$$

- ▶ (Optional) Damping: $c^{(i+1)} \leftarrow (1 - \alpha)c^{(i)} + \alpha c^{(i+1)}$, $\alpha \in (0, 1]$.

- ▶ Set $k'_{m,n} = R \bar{k}_m + \bar{w}_n - c_{m,n}^{(i+1)}$

4. Convergence:

- ▶ If $\max_{m,n} |c_{m,n}^{(i+1)} - c_{m,n}^{(i)}| < \varepsilon$, stop.

- ▶ Otherwise, form $\pi^{(i+1)}(k, w; \bar{\mathbf{k}}, \bar{\mathbf{w}}, \mathbf{c}^{(i+1)})$ (your interpolation rule) and continue.

Forward rollout vs TI and EGM

- ▶ Versus TI: TI is implicit (solve a root/NCP at each node with $k' = R\bar{k}_m + \bar{w}_n - c_{m,n}^{(i+1)}$). Forward rollout is explicit (uses $k'^{(i)}$ from $c^{(i)}$). Same fixed point if it converges; usually needs damping.
- ▶ Versus EGM: EGM fixes a k' grid, inverts Euler to current k , interpolates to the k -grid, enforces the cap on the lower tail. Forward rollout stays on the current $(\bar{k}, \bar{w})$ grid; no back-mapping stage.
- ▶ Pros: cheaper per iteration; nodewise decoupled and parallel; only interpolate $\pi^{(i)}(k', \bar{w}_{n'})$ in k' .
- ▶ Cons: may oscillate/diverge when $\beta R \approx 1$, with flatter u_c (small σ ; large intertemporal elasticity of substitution), or near kinks., or near kinks; Euler holds only in the limit; always re-impose $c_{m,n}^{(i+1)} \leq \bar{\pi}_{m,n}$.